

3	(a)	Any 2 of the following: - Pressure - Average kinetic energy - Root mean square speed - Internal Energy		A1
	(b)	(i)	$pV = nRT$ $n = \frac{1.0 \times 10^{-2} \times 300 \times 10^3}{8.31 \times 290}$ $= 1.24 \text{ mol}$	M1 A1
		(ii)	Mass of 1 mole = $4.82 \times 10^{-26} \times 6.02 \times 10^{23}$ $= 2.90 \times 10^{-2}$ Mass of air = $1.24 \times 2.9 \times 10^{-2} = 3.61 \times 10^{-2} \text{ kg}$	M1 A1
		(iii)	$\rho = \frac{3.61 \times 10^{-2} \text{ kg}}{1.0 \times 10^{-2}}$ $= 3.61 \text{ kg m}^{-3}$	A1
	(c)	(i)	The <u>increase in the internal energy of a system</u> is equal to the <u>sum of the thermal energy added to the system by heating and the work done on the system.</u>	B1 B1
		(ii)	The specific heat capacity of a substance is the thermal energy required to raise the temperature of unit mass (1 kg) of the substance by 1 kelvin (K).	B1
		(iii)	At constant volume, the gas does not do any work thus the thermal energy required to raise the temperature of unit mass of the gas by one kelvin is only the increase in its internal energy. At constant pressure, the thermal energy required to raise the temperature of unit mass of the gas by one kelvin is the sum of the increase in its internal energy and work done by the expanding gas, where work done $W = p\Delta V$. Thus the specific heat capacity at constant pressure is greater than specific heat capacity at constant volume, with the difference being due to the amount work done by expanding gas.	B1 B1 B1